

TABLE OF CONTENTS

I-005 DISTRIBUTED CONTROL SYSTEM

	Page No.
1 Scope of Work.....	I 005-3
2 Codes and Standards.....	I 005-4
3 Environmental Requirements	I 005-6
4 Design and Operating Conditions	I 005-6
5 Detailed Requirements.....	I 005-6
5.1 General	I 005-6
5.2 Power Failure.....	I 005-8
5.3 System Architecture	I 005-8
5.3.1 General.....	I 005-8
5.3.2 Process Automation System	I 005-9
5.3.3 Plant Management System.....	I 005-10
5.3.4 Engineering and Diagnostic system	I 005-10
5.3.5 Plant Control and Monitoring	I 005-10
5.3.6 Communication System.....	I 005-11
5.4 Specification of System Components	I 005-11
5.4.1 Electronic Cubicles.....	I 005-11
5.4.2 Marshalling Boards	I 005-11
5.4.3 Processing Units	I 005-12
5.4.4 Interface Module (for PLC)	I 005-14
5.4.5 I/O Modules and Signal Conditioning.....	I 005-15
5.4.6 Process Interface	I 005-17
5.5 Functions	I 005-19
5.5.1 Data Acquisition.....	I 005-19
5.5.2 Binary Control.....	I 005-19
5.5.3 Analog Control.....	I 005-20
5.5.4 Historian.....	I 005-21
5.5.5 Interfaces to PLC.....	I 005-22
5.6 Plant Bus.....	I 005-22
5.7 Process Operating and Monitoring	I 005-23

5.7.1	Work Stations.....	I 005-24
5.7.2	Operator Keyboard	I 005-29
5.8	Plant Management System	I 005-30
5.9	Engineer and Diagnostic System	I 005-30
5.9.1	Engineer Workstation	I 005-32
5.9.2	Engineer System.....	I 005-33
5.9.3	Diagnostic System.....	I 005-34
5.9.4	Portable Engineer Station (for PLC)	I 005-35
5.10	Power Supply and Grounding.....	I 005-36
5.11	Application Software	I 005-36
5.11.1	Programming Facilities.....	I 005-38
5.11.2	Programming Methods	I 005-38
5.11.3	Hierarchical Structure and Standard Functions.....	I 005-39
6	Testing.....	I 005-39
6.1	Factory Testing	I 005-39
6.2	Testing at Site	I 005-39
6.3	Test Equipment	I 005-42
7	Warranty.....	I 005-42
8	Quality Assurance / Quality Control (QA/QC)	I 005-42
9	Spare Parts.....	I 005-43

I 005 DISTRIBUTED CONTROL SYSTEM

1. SCOPE OF WORK

The scope of this specification is to provide the complete hardware and software of a microprocessor based control system required to form an integrated functionally distributed control system (DCS), the parts of which are linked by a high performance redundant data transmission system.

The work to be performed under this technical specification shall include the design, manufacture, inspection and testing at the manufacturer's facility, packing for export, shipment, clearance through customs, delivery to site, erection, testing, commissioning, warranty and inspection prior to final acceptance.

The plant control system as the integration of several programmable microprocessor based control systems consists of five main operating groups:

- Processing Automation Units:
 - a) Signal Conditioning and pre-processing
 - b) Data transmission
 - c) Binary control
 - d) Analog control
 - e) Protection functions
- Man-machine interface for supervision and control
- Engineering, diagnostic and service tools
- Plant Management System
- Historian System

The basic design shall have the capability to be interfaced with other control systems and to be synchronized with the plant master clock.

The process control system (including the engineering workstation) shall be based on "open" system architecture using Object Linking and Embedding for Process Control (OPC). The system shall have inherent capability to integrate and exchange information with other brand system devices and platforms via industrial standard communications platforms and protocol. Time tagging resolution for individual event data shall be as set out in the individual functional descriptions.

Several process controllers shall be able to be interconnected, and together with the central operator and plant management stations, make up a control network. Several control networks shall be able to be interconnected into a complete plant network, which can share centrally located operator, engineering and plant management stations.

The contractor shall implement the functional requirements of the above operating groups through the DCS.

The plant control systems shall be designed and provided to perform start up, routine loading operations and for emergency plant shut down from the Central Control Room (CCR). The bidder shall provide in his proposal a detailed description of the plant control systems to be based on a Distributed Control System.

The 500 kV substation control system is not part of these specifications and shall have its requirements as specified in Section 3 Technical Specifications E-460 "Substation Control System". The two systems shall, however be linked on the control network.

2. CODES AND STANDARDS

The design, manufacturing and testing of the system shall comply with the IEC code and standard or other reputable standard.

No purchasing fabrication or manufacture shall be proceeding prior to the approval of the accepted Code and Standard.

ANSI ASCII 1	American Standard Code for Information Interchange
IEC 60050	International Electrotechnical Commission
IEC 60051	Direct acting indicating electrical measuring instrument and their accessories
IEC 60073	Colours of indicator lights and push-buttons
IEC 60083	Plugs and socket-outlets for domestic and similar general use
IEC 60113	Diagrams, charts and tables
IEC 60617	Recommended graphical symbols
IEC 60144	Degrees of protection of enclosures for low-voltage switchgear and control gear
IEC 60146	Semiconductor convertors
IEC 60947	Low-switchgear and control gear
IEC 60228	Conductors of insulated cables

IEC 60245	Rubber insulated cables of rated voltages up to and including 450/ 750 V
IEC 60255	Electrical relays
IEC 60269	Low voltage fuses
IEC 60309	Plugs, socket outlets and couplers for industrial purposes
IEC 60331	Fire-resisting characteristics of electrical cables
IEC 60445	Identification of equipment terminals and of certain designated contractors, including general rules for an alphanumeric system
IEC 60446	Identification of conductors by colours or numeric.
IEC 61204	Low-voltage, power supplies, DC output
IEC 60529	Classification of degrees of protection provided by enclosures
IEC 61124	Reliability testing – compliance test for constant failure rate and constant failure intensity
IEC 61131	Programmable Controllers (all parts)
IEC 61508	Safety Integrity
IEEE C 37.90.1	Guide for surge withstand capability (SWC) test
ISA S5.1	Instrument Symbols and Identification
ISA S5.2	Binary Logic Diagrams for Process Operations
ISAS5.3	Graphic Symbols for Distributed Control/Shared Display Instrumentation
ISA S50.1	Compatibility of Analog Signals for Electronic Industrial Process Instruments
NFPA 70	National Electrical Code
NEMA CC 1	Electrical Power Connectors
NEMA WC 30	Color coding of wires and cables
SAMA PMC 22.1	Functional Diagramming of Instrument and Control Systems
SAMA PMC33.1 OR IEC 801/3	Electromagnetic Susceptibility of Process Control Instrumentation.

3. ENVIRONMENTAL REQUIREMENTS

The equipment furnished under this Technical Specification shall be suitable for operation in a dual gas and oil fired combined cycle power plant in a tropical climate near to the coast.

Equipment shall be able to withstand, but not limited to the following operation condition:

- Temperature Maximum 50°C
- Humidity Maximum 90%
- EMI/RFI 1.2 m away from 5 watts, RFI frequency up to 470 Mhz

The Contractor shall arrange and calculate its requirement for the air conditioning system based on the equipments' ideal operating conditions.

4. DESIGN AND OPERATING CONDITIONS

A plant wide automation based on microprocessor control systems designed for a combined cycle power plant shall be supplied. The process automation shall be performed through a functionally Distributed Control System (DCS) which shall include processors, I/O modules and operator interfaces.

All Control Modules and I/O cards shall be housed as individual modules in suitable cabinets with front and rear access. Cabinet environment shall be maintained with the following:

- Dual redundant fans to provide good air circulation inside the cabinet
- Temperature conditions inside the cabinet shall be continuously checked by a built-in temperature sensor
- Cabinet fans and temperature status shall be checked continuously by the system self diagnostics and shown to the operators in the system diagnostic displays.

5. DETAILED REQUIREMENTS

5.1 General

The DCS shall be in accordance with Book I Part 1 Clause 8.5.2 Item c and assembled in Indonesia. The DCS shall fulfill all requirements of the power plant operation, supervision, and maintenance, shall optimize operational displays, and shall have the following features:

The Plant Automatic Systems shall be organized into hierarchical levels for clear operation and supervision of different control tasks:

- Individual control level I/O, local and protection
- Unit control level for plant control
- Plant Coordinated control level for load control and APS (Automatic Plant Start-up and shutdown)

- Unit control level for operator interaction
- Plant management level for production data calculation and presentation

The Plant Automation Systems shall be supervised and operated by Personal Computer (PC) based operator stations in the Central Control Room.

As far as possible, all plant shall be directly controlled by the DCS including proprietary Balance of Plant (BoP) items. Other ancillary systems shall be integrated via serial/parallel link to the DCS.

The Plant Automation System shall provide functional separation to avoid one functional group interfering with the operation of any other group or any other group on other levels.

The Plant Automation Systems shall be designed to enable expansion from a simple process control to a wide range of project applications to form an integrated control system. The expansion shall be able to be executed on-line without any disruption either to the process or communication system.

The contractor shall be responsible for overall design; factory test, supply, installation, site test, and commissioning of microprocessor based Distributed Control System (DCS), including all instrumentation and control equipment in accordance with the bid drawings and as required under other sections of this Technical Specifications.

The Contractor shall also provide information and monitoring equipment and its programming to adequately inform the operator on the plant status, and to alert the operator to out of limit conditions in equipment or process.

The system shall be of proven design, backed up by experience in installation of similar size with successful operation for at least 12 (twelve) months. The proposed equipment shall be able to be serviced from a technical service office in Indonesia.

Automatic protection and power plant trips shall be provided in order to maintain the power plant always in a safe and un-damaged condition without any operator involvement.

The Contractor shall provide all control system configuration and all software required to enable the system to perform its specified functions. The software shall include a Real-Time operating system for each node or drop and all application software required to meet the specified functional, operating and general system requirements.

The Operator Stations shall be easily connected to the Plant Bus. The Operator Stations shall offer advanced functions for color-graphic displays, manual control operations, event and alarm handling, trend curve presentation etc.

High availability of the DCS shall be achieved by using redundant processor modules, redundant voltage regulators and mains supplies.

MEMO
-029

Controller DCS shall be fully modular. The subracks are normally installed in cabinets. For field wiring requirements see Part 4 Section 3 Technical Specifications I-001 "Instrumentation and Control".

The design shall implement remote I/O as extensively as possible to reduce hard wire from the field to the Electronic Equipment Room and numbers of the marshalling cubicles in the Electronic Equipment Room. Except for transmitter and analyzer electronic modules, no other electronic modules shall be installed in the field.

Several process control systems may not be controlled directly by the DCS but they shall be monitored from the DCS, with sent/received signals to/from DCS for operation and alarm purposes. These applications shall include all necessary control, protection, alarm as well as operating and supervisory functions and the related hardware Process Automation Units.

The following process control systems may be controlled by external systems:

- Turbine Supervisory Systems.
- Turbine Protection Systems.
- Generator Protection Systems.

These external systems shall generally be connected to the Plant Management system of the DCS by a redundant point to point bus (Gate Way) using fiber optic cables.

The necessary space for cubicles of these systems shall be considered in the layout of the Electronic Equipment Room.

5.2 Power Failure

In the event of a power failure, the central processors shall be capable of gradual degradation, any necessary values shall be preserved to permit an orderly restart, and the computer system shall shut down without corruption. An automatic restart feature shall be incorporated. Time shall be automatically reset by the time signal from the satellite communication.

5.3 System Architecture

5.3.1 General

The Plant Automation Systems shall be designed as an integrated management and control system and shall provide plant performance calculation for major plant devices and optimized schedules for plant management.

The Plant Automation Systems shall be in a stable control while on-line plant performance calculation is being executed.

5.3.2 Process Automation System

This section defines the functional requirements of control capability such as algorithms, loading requirements and guidelines on architecture. The bidder shall propose a system that fits the specification requirements.

The control capability shall be made available for modulating control through preprogrammed algorithms and easy fill-in the blank type entry forms which is understandable by instrument engineers without the ability of programming experience.

All control and monitoring functions including analog, sequencing, batch, switches, digital I/O and etc, shall be executed within the same type of controller modules. There shall be no separate controllers for sequence and analog control. Signals listed in the I/O list for each group in the plant shall be distributed equally among all control modules assigned for each plant.

Power failure to a control module for less than 72 hours shall not require the down load of the control module. Rechargeable batteries for 3 days minimum shall back up the contents at of the memories of the Control Modules

The bidder shall provide 10% spare installed control modules and provision for the 20% space for future extension.

Remote I/O capability shall be available as a standard product.

Remotely located I/O cards shall be interfaced via redundant twisted-pair cables. Redundant fiber optic cables shall be used when I/O cards are mounted in other buildings far from the Control Module. The communication speed of fiber optic cable used for remote I/O shall not less than 10 Mbytes per second.

Each unit configuration shall provide sufficient backup for critical subsystems. Transfer or switch over to the backup shall be automatic and bumpless without any disturbance to the plant control. Subsystem malfunctions shall be indicated and where backup subsystems are provided.

In sizing the system, the Bidder shall consider the expansion capability of process and the system loading in accordance with the response time.

The control module and input module shall be able to handle a wide variety of signals including analog and digital.

Data transmission between the electronic cubicles in the electronic rooms, central control room and remote I/O shall be performed through a redundant Plant Bus System. The Plant bus shall be designed with minimum speed of 100 Mbytes per second.

Data transmission between Processing Unit shall be performed through an Ethernet LAN. The plant network shall be designed with TCP/IP based and speed of 1 Gigabytes per second.

On line connection or disconnection of either LAN or a processing unit of the DCS shall not affect the running communications systems.

5.3.3 Plant Management System

The plant management system is a part of the overall Plant Automation Systems that shall be operated from a PC based operator console located in the Plant Manager's office. The communication between the Plant Management Station and the processing unit of the DCS shall be performed via the Plant Bus.

The detail of the plant management systems is given in Part 4 Section 3 I-200 "Plant Management System"

One heavy-duty log printer and one hard copy unit shall be provided to the plant management system.

5.3.4 Engineering and Diagnostic system

The engineering and diagnostic system is a part of the overall plant control system that shall be operated from a PC based work stations located in the Plant Engineer's Room. The communication between the work station and the processing unit shall be performed via the Plant Bus.

The engineering system shall be designed as a single engineering system that enables multiple users to access the system configuration with a uniform user graphic interface.

The engineering system shall be able to perform all detailed engineering for commissioning, modifications and updating.

One heavy-duty log printer and one hard copy unit shall be provided for the engineering and diagnostic systems.

The engineering and diagnostic systems shall be operated from a dedicated engineering station or from any available operator station (password protected) in the Central Control Room.

5.3.5 Plant Control and Monitoring

The process and monitoring station shall provide powerful equipments for plant control and monitoring, for communication networking (including internal network in the control room) and for access to explore and analyze plant data to the other databases through an internal network and/or Plant Bus.

The plant control and monitoring station shall be emphasized on efficiency, ergonomics and process security of the operation by providing all of the operation function such as alarming, reporting, logging and other operator process interface through flexible, dynamic, interactive color graphic displays.

The plant control and monitoring station shall enable real time process information exchange and operations directly from the plant management system or Load Dispatching Center while performing the operator's process interface.

The plant control and monitoring shall facilitate automatic checking of interlocks at each step and provide for continuous monitoring of operating limits and tripping conditions. The Plant Bus shall be able to integrate plant data net work.

5.3.6 Communication System

The Plant Bus shall be designed redundantly based on fiber optic technology and connected between Central Control Room, Electronic Equipment Room and the other Electronic Room, such as GT Local Control Package.

The Plant Bus shall be designed in industrial standard protocols according to ISO/OSI. No single fault element shall disrupt communication system. The faults system shall generate a diagnostic message to the operator station.

The Plant Bus connecting between buildings, such as between Electronic Equipment Room and GT Local Control Package shall be Fiber optic cable.

Cable for serial data link between PLC and DCS shall be fiber optic cable.

The information data for printers in the central control rooms shall be designed with the different data bus (LAN) from Plant Bus.

5.4 Specification of System Components

5.4.1 Electronic Cubicles

The electronic components of the DCS shall be installed in closed cubicles located air-conditioned electronic equipment rooms.

The modules of CPU, memory and interfaces shall be on racks and to be separated from the power supply units and I/O modules.

For extension purposes the electronic cubicles shall have a minimum reserve space for 20% hardware modules.

The controller modules shall be possible to be replaced without interfering with the external wiring.

5.4.2 Marshalling Boards

All process measuring and command loops shall be cabled to the field side of the terminal boards and shall be cross-wired to the second terminal row which is connected by prefabricated cables to the DCS input/output modules.

For extension purposes the marshalling board shall have a minimum 15% spare for terminals and cables connections.

5.4.3 Processing Units

Processing unit as part of the DCS shall cover all control, regulation and protection functions, transfer commands from the operation and monitoring stations to the Plant Automation Systems through the process data requested by engineering and diagnostic station to the operation and monitoring station. It is preferred all processing units have similar type of controllers.

The control module of DCS shall be redundant fault tolerant (FT) processor modules, each module containing CPU, memory and switch over function. A dual fault tolerance system for control processor, high-speed data high way, and critical I/O points shall be implemented to perform a real time information transfer between the various DCS components.

The processor as fault tolerant provide continuous operation, of the controller in the event of virtually any hardware failure within one module of the pair, both modules receive and process information simultaneously and fault detected by the modules themselves.

The plant control systems redundancy shall be designed in such way that neither hardware nor a software failure will result in the loss of single process control system. The plant control systems shall be supported with system navigation tools

When a module, card, or power supply fails, the failed redundant module, card, or power supply shall be able to be replaced on line without system shut down or operator interaction to system such reboot. Likewise, additional I/O module, controller and HMI shall be done online.

Microprocessor shall be 32 bits type on its operating system or better. It shall be a high power computing processor. The control modules shall be non-obsolence devices.

Each microprocessor based module shall be capable of accepting unconditioned inputs from the field via a termination cubicle. Modules provided with a single microprocessor shall be provided with a back-up battery to hold the information during power loss. On restoration of the power supply, the microprocessor shall up date status and be ready to resume operation.

Each controller or microprocessor furnished under this specification shall be the proven latest technology and reliable in operation.

The CPUs shall provide internal fault analysis with predetermined fail-safe failure mode in the event of a fault. An external contact shall indicate a CPU fault.

In case of power supply interruption, automatic restart of the CPU shall be provided without operator interference.

The CPU load conditions shall be as follows:

- Peak load condition: less than 60% of available capacity
- Normal load conditions: less than 40% of available capacity

The processing time of particular controller shall be dedicated to the requirement of controlled technological process and maximum processing time including input operation, logic or function processing and output operation shall not exceed the following limits

- Closed loop control: 250 msec
- Open loop control: 150 msec
- The processing unit shall provide all timing requirements as follows:
- The refresh frequency of any indication shall not be less than once per second.
- The time between the change of state at a system input module of discrete status input and the implementation of the change within the system shall not exceed 0.1 second.
- The time between the changed of state at a system input module of general alarm input or between the generation of such an alarm within the systems and the initiation of an alarm message or output shall not exceed 0.01 second.
- The resolution of alarms in the sequence of events recorders group shall not exceeds 0.001 second.
- The time between the generation of report for any analog value and its implementation within the system shall not exceed 0.1 second.

Modulating control functions shall be represented by software blocks. The tags in the modulating control shall be able to access values from and send values to other points in the modulating controls within the same Controller Module or other Modules on the network.

Without custom configuration the basic PID control algorithm shall be based on velocity calculation type and the tracking of cascade loops shall be made automatically so that bumpless transfer to cascade mode can be achieved at any instant of time without preparing the configuration for signal tracking. When a controller in the cascade hierarchy is turned to cascade open or to manual mode, the upper level controllers (primary controllers) shall sense the mode change of the controller. The output signals of the upper level controllers shall track the set point of each controller at the next lower level automatically.

Without custom configuration, bad data status shall be propagated among cascaded loops.

Bad data status shall be triggered as a result of the following conditions:

- Losing communication with I/O Module
- Loss of signal with field devices
- I/O Module failure
- Invalid calculation.

Adaptive type PID controllers capable of automatically predicting and setting tuning parameters shall be available in the system. Discrete tag number control functions are required at the modulating level for handling discrete points such as digital input, digital output, timer, logic gate etc.

Flags shall be available to be used and triggered by sequence functions or manipulated by the operator manually.

Switch block shall be available with the capability to support the control and operation of ON/OFF valves and other devices. The switch shall initiate alarm to the operator when feedback timeout is detected. Timeout shall be configurable within 0 to 1000 seconds with 1- second resolution. This shall be achievable without configuring external timers to support this function.

Calculation shall be provided in the form of blocks supporting modulating and sequence control function as well as providing general-purpose calculation function.

5.4.4 Interface Module (for PLC)

Interface to other systems/devices shall be provided completely operational including all necessary hardware and software.

The system shall be able to support various types of field bus.

The control network shall be based on TCP/IP and the control network utilized IEEE 802.3. The following interfaces shall be provided but not be limited to:

- Gate Way interface

- In case of particular controller is part of the plant control system, the complete duplicated Gate Way interface shall be provided for signal exchange with the communication speeds of 10 Mbit/sec.
- RS 232C / RS 485 interface
- The standard RS232C / RS 485 interface module shall generally be provided including all necessary hardware and software with “D” type connector.

5.4.5 I/O Modules and Signal Conditioning

The following clauses cover both local and remote I/O. Both local and remote I/O modules shall be able to be configured from the engineering station in the Central Control Room and shall have built-in diagnostic that enable to be displayed on the Engineering and Diagnostic stations.

In the event of communication failure the remote I/O modules shall be able to be configured locally. The DCS shall hold the previous value or to go to a predetermined value.

All inputs and outputs shall be isolated from the internally used circuits. Each isolator shall be replaceable on-line without disconnecting the wiring between the DCS and field instrument, and without affecting other signals in the system.

If intrinsic safety signals are specified then active barriers shall be quoted as alternative to the DCS isolators.

All I/O shall be isolated from the others and from ground in order to avoid erroneous signals caused by ground loops.

The mechanical components of the I/O Module shall be characterized by a highly modularized design with four basic parts; processor modules, the field communication unit, I/O modules and field wiring termination units, the module termination modules, which act as I/O module carriers. The processor modules and field wiring termination units shall be mounted on rails.

The I/O modules shall be protected by a mechanical keying arrangement from being inserted into the wrong position.

Hot-swap of I/O modules allowing replacement of faulty modules without disconnecting field power of system power to the I/O station shall be possible. The I/O module assembly shall be plug-in type to enable on-line replacement without affecting field wiring.

All analog and binary signals shall be connected to input modules for signal conditioning and data transfer to the main bus system. The I/O modules shall include all necessary shielding, filtering and noise rejection circuitry, high and low level internal signal isolation, analog to digital conversion to provide a

stable digital signals for control, display calculation and data storage purposes consistent with system accuracy requirements.

The following types of I/O shall be provided:

(i) *Analogue Inputs*

- The analogue input cards and modules shall be capable of receiving 4~20 mA dc signals. The analogue input circuitry shall be capable of providing a +24V dc voltage, current limited source for powering associated two-wire transmitters. Minimum resolution of the Analog to Digital conversion shall be 12 bits.
- Inputs shall be supervised for live-zero, which means that an alarm will occur if the input is less than 3.9 mA or higher than 20.1 mA.

(ii) *Analogue Outputs*

- Isolated analogue outputs of 4~20 mA dc shall be provided for the field devices. They shall be driven from the control system also directly from application programs in the high level procedural language. The controller output directions shall be configurable, i.e. direct or reverse acting.
- Minimum resolution of the Analog to Digital conversion shall be 12 bits.
- The outputs shall be short circuit protected.

(iii) *Thermocouple and RTD*

- The system shall be capable of accepting all standard thermocouples and RTDs. Thermocouple cold junction compensation and open circuit detection shall be provided and reported at the work station.

(iv) *Millivolt Inputs*

- The system shall be capable of accepting low level millivolt signals from other devices.

(v) *Pulse Count Inputs*

- The system shall accept pulse signals from external devices, the total accumulated pulses being available to the control database as a process variable. Pulse count inputs shall be 24 V dc internally powered.

(vi) *Digital Inputs*

- The system shall accept digital (on/off) inputs for process alarm conditions or status. Digital inputs shall be dry contact type connected directly to transistor type or thyristor operated at 24 V dc or 240 V ac and may be internally or externally powered. The 240V ac shall only be connected into interposing cubicles, NOT directly into the control system. All inputs shall be optically isolated with the

change of status reported at the work station and available to logic application programs. The input modules shall also be provided with over voltage protection.

- Power supply to a group of buffer relays (32 maximum) shall be protected by replaceable fuse. Digital input cards shall have software configurable filter to eliminate contact chattering.
- The Owner has a strong preference for solid state proximity probes, with all field binary devices supplied for control system, where supply fault monitored by the control system
- Rated input current shall not less than 5 mA (24 VDC) so that
 - a. -30 VDC to +4 VDC correspond to binary “0”
 - b. +13 VDC to +30 VDC correspond to binary “1”

(vii) *Digital Outputs*

- Digital outputs shall be 24 V dc or 240 V ac externally powered. All outputs shall be capable of being:
 - a. momentarily pulsed “on”
 - b. off command latched on until an “off” feedback signal is received
 - c. pulsed “on” for a predetermined variable time period.
- They shall also be provided with over voltage protection for induced surge voltage caused by the switching of inductive loads.
- It shall be possible to set the outputs to a predetermined state “on” or “off” on power up or power down. It shall be possible to set or reset the digital outputs from the operator’s workstation via the LAN and by Logic Application programs.
- Rated output current shall not less than 5 mA (24 VDC)

(viii) *BCD Inputs and Outputs*

- The system shall (optionally) accept BCD inputs and outputs up to five decades. Inputs may either be contact type or solid-state. Outputs shall be solid-state.
- 20 % spare each type or kind of I/O module shall be provided.

5.4.6 Process Interface

The system shall contain process interface (I/O) hardware to provide data acquisition, and outputs to valves, actuators and discrete devices.

The system shall be capable of connecting to other I/O sub-systems, which may be fieldbus (Profibus, ControlNet, DeviceNet, AS-I or Fieldbus

foundation). The system shall also have the capability of supporting network configurations (Ethernet and network protocols, Net Bios and TCP/IP), with NON proprietary gateways or custom protocols enabling full connectivity to LANs, WANs, Internet, Intranet maintenance management tools.

The interface hardware shall be capable of performing as follows:

a. Inputs

- Scan, process, filter and convert to engineering units, linear analogue inputs such as transmitter current signals and field transducer voltage signals.
- Scan, process, filter, linearise and convert to engineering units, non-linear analogue inputs such as thermocouples, RTDs and analyser inputs.
- Simulating analogue and binary I/O without changing any logic configuration with indication of simulation indicated on the engineering tool/alarm system.
- Provide square root functions for flow devices (mass flow computation).
- Provide alarm detection with a time tag for all inputs to acceptable sequence of events resolution.
- Detect change of state with time tag of discrete inputs to acceptable sequence of events resolution of 5 mS.

b. Outputs

- Send continuous domain scaled control signals to valves and actuators.
- Send digital outputs to actuators under Logic Application Program control.

c. General

- Communicate the current values of all inputs, analogue and discrete, and display them at any operator's work station.
- Communicate with PLCs and display all or any selected data on the operator work station.
- Perform automatic diagnostic testing to identify failures and errors and report such failures via the operator's work stations.
- All I/O modules shall withstand a continuous series mode voltage of 240 Vac rms without damage (other than fuse rupture, if used).
- Configure, set, read and modify analogue controllers and discrete I/O's through the work station without having to connect a separate terminal to the Local Area Network (LAN).

- Download data to the system's equipment connected to the LAN from the work station.
- Provide appropriate series mode and common mode noise and RFI rejection.

5.5 Functions

5.5.1 Data Acquisition

Details of data acquisition, scanning and alarming are carried out as part of the Plant Management System. See Part 4 Section 3 I-200 "Plant Management System"

5.5.2 Binary Control

This hardware shall be capable of receiving binary signals from the process, apply Boolean logic and output resulting signals to the final elements such as motor starters, switchgear, solenoid valve, etc.

The binary control, as part of the DCS shall be designed with three different hierarchical levels. The lower level contains the drive control, the middle level contains group control and the highest level contains unit control.

The connection between the drive control system and switchgear equipment shall be operated by 24VDC or 48 VDC command and feedback signals initiating suitable relays which shall be located in the inside of the MCC itself or in the cubicle in the switchgear room.

a) Drive Control Level

The drive control function shall cover the control function for motors, motorized valves, solenoid valves and other components with binary control characteristics.

The drive control function shall have the following features as minimum:

- No active signals to the switchgear shall be sent from a faulty drive control function.
- Interlock and control loops shall be located on the intelligent drive control module if necessary.
- All protective signals to be displayed on the operator station.
- The protection trip signal shall have precedence over any other commands and the protection trip signal shall block the ON commands or OFF commands until they are acknowledged, if necessary or specified.
- Blocking of "pumping" circuit breakers in the presence of both ON and OFF commands.

- Supervision function and alarm for:
 - Status discrepancy.
 - Motor control center not available.
 - Drive control, unit fault.
 - Running time exceeded.
 - Motor over current.

b) Function Group Control Level.

The task of the function group control system is to control the drive controls that are allocated to a process function group by means of a fixed program. The drive controls shall be controlled based on events or correct chronological order to fulfill the process requirement of the related function group control. The drive control in hierarchical structure is a subordinate of the function group control level.

A drive group control is not defined according to process function, but from specific aspect of the drive.

Regarding the generation of the command signals there are two basic structure of function group control system:

- Combined control (logic control), where the output signals are formed by the logically combined input signals.
- Sequential control, in a step-by-step sequence program, each step shall only be executed when the predetermined conditions are fulfilled.

The programming system of the function group control shall be divided in accordance with the process requirements into the following sub-programmers.

- Start-up program.
- Operating program.
- Shutdown program.

Each function group shall have ON, OFF and auto/manual or stop function via VDU based Operator Station. The VDU shall include indications for ON, OFF AUTO/MANUAL or STOP and FAULT status.

5.5.3 Analog Control

This hardware shall be capable of receiving analogue signals from the process, apply single and multi variable loop control algorithms and output the resulting control signals to the final control elements. In the event of plant bus

failure, the hardware (controller) shall be stand alone and capable of performing all the control functions. The configuration data shall be downloaded from the work station over the plant bus. The controller memory shall be non-volatile. The configurable algorithms shall be as required to meet the process control requirements.

The analogue control loops shall be as set out in Part 4 Section 3 I 100 “Plant Control System”:

The analog control system shall be designed in a hierarchical manner with two levels of hierarchy. The lower level includes the sub-controller, whereas the master controllers form the higher level. If a master controller fails the subordinated based controller shall be switched to remote manual control.

Most of the analogue control functions for the combined cycle plant shall consist of dedicated single loop controls that shall be included in the DCS.

Control and monitoring of each controller shall be possible from the Operator Station.

Considering the above requirements the analog control shall have at least the following features:

- The controller shall be equipped with the necessary software functions to fulfill all of the typical control tasks such as I, P, PI, PD, PID controller, function generator, parameter selection adaptive control.
- Automatic and manual control change over shall be bumpless transferred.
- The controller shall be able to receive signals from the sequential control system through Plant Bus.
- During and after commissioning phase, the control structure and the control parameters of the analog control shall easily be modified on-line by engineering tool.
- Any faults in the automatic control shall switch the controller to the manual control. The operating and monitoring system shall guide the operator via the overview and group display to determine the cause of fault of the faulty controller.
- The controller shall provide anti-reset wind-up.
- Under no circumstances should transmitter deviation or large process disturbances force the controller to manual. The preferred reaction is to block the controllers output or apply the appropriate, predetermined protection command.

5.5.4 Historian

Hardware and software for archiving data shall be included. Details of the requirements for this are given in Part 4 Section 3 I-200 “Plant Management System”

5.5.5 Interfaces to PLC

All serial links to connect to third party PLCs and other electronic equipment shall be supplied as required. Whenever possible, connections shall be directly to the Plant Bus (i.e. be individual nodes on the loop).

This Control Module shall have an open standard serial protocol for communication with other manufacturers control system (PLCs). The Controller shall be able to act as a master or slave station.

The interface software shall provide interfacing facility to common protocols of the PLC such as Modbus protocol or Ethernet. The PLC interface shall be an integral part of the DCS.

The PLC systems shall be similar type and brand name for the whole plant. If the PLC is supplied by the DCS vendor, the PLC's programming language should be similar to the DCS's programming language to have an integrated and seamless control system.

5.6 Plant Bus

The Contractor shall provide a redundant Plant Bus communication/networking system between different parts of the Instrumentation & Control system. This communication bus shall connect all parts of the I&C system (whether physically distributed or not) including processors, controllers, I/O facilities and other peripheral equipment. The Plant Bus shall be designed for a combined cycle power plant with minimum speed of 100 Mbits per second.

The Data Communications System shall be able to use a wide variety of communication protocols, each one specifically selected herein to optimally perform the specific communication task required.

TCP/IP is a widely accepted communication protocol, used at the plant network level for file transfers and client server database transactions between both internal and external stations. Transmission speed is minimum 1 Gigabits/s.

Plant Bus is an IEEE 802.3 communication bus used in plant and control networks to handle high data transmission rates at 100 Mbits/s. Control bus /Field bus interconnects Controllers to various drive systems. Then stand-alone, peer-to-peer or subordinate process control networks are connected in a highly deterministic way at 2Mbits/s.

Multi vendor Interface is a plug-in board for Controllers, allowing serial communication with other control system with many standard protocols available.

Gate Way Network Communication is available for different requirements for high performance in large systems or / and also different bus alternative are functionally compatible, and can replace each other with no effects on

application programs, as long as the resulting configuration is in accordance with the Gate Way Network configuration rules.

The communication system shall be accomplished through digital communication bus providing reliable and high-speed communication between interconnected devices.

The Plant Bus communication hardware and software will be such that complete failure of one node shall not have any effect on communication between other nodes. Loss of communication shall not cause loss of control at the processing unit. Loss of a processing unit shall not disrupt communication to other processing units.

To enhance the availability/reliability of the plant control system a comprehensive error checking shall be performed for both channels on-line actively and cyclically to assure message integrity.

On-line connection or disconnection of either LAN or a processing unit of the DCS shall not affect the running communication system.

5.7 Process Operating and Monitoring

The operator human-machine interface shall consist of redundant PC based operator consoles for supervision and operation of the power plant process in automatic and manual mode. The operator interface shall be by way of VDU/keyboards through which the operator will be able to receive information on the status of the plant and control changes to the plant.

Access to the Operator Station shall be provided by a multi-level password system.

The Operator Station VDU shall be provided with the following requirements as minimum:

- 21" TFT-LCD colour monitor.
- 1280 X 1024 pixel resolution.
- 256 distinct colours.
- brightness and contrast adjustment
- Screen build time within 2 (two) seconds.
- On-line screen refresh within 1 (one) second.
- Display 48 lines with minimum 80 characters per line.
- Adjustment density.
- Digital display adjustment

At least one high speed text and one colour graphics printer shall be provided for alarm logging and graphics screen printing to be located in the central control room. The printers shall be free standing with a noise level not to exceed 70 dB(A) at a distance of 0.5m. If the noise level exceeds the level nominated then the printers shall be housed in an acoustic enclosure. The printer interface shall be IEEE-488, RS-232 or Centronics parallel connection. The printers shall include paper trays.

The operator station printer shall have the following requirements as minimum:

- The print speed shall be 150 pages per minute for laser jet and 140 characters per second for dot matrix printers dedicated for data logging.
- Low paper sensor.
- Adjustment forms tractor and from feed control.

5.7.1 Work Stations

The DCS system for power plants shall be based on the proven latest open automation platform and having Open Control System hardware platform in which support and provides the Dynamic Data Exchange (DDE) and OLE for Process Control (OPC) interface.

The operator console shall consist of ergonomic furniture to contain TFT-LCD screen and one large screen operator work stations, emergency stop pushbuttons for the GT and ST, telephone, pager and audible alarm

A Windows ~~or UNIX~~ based HMI shall be provided. The open universal workstation shall share data with the open History Station MIS to enable multi-dimension information exchange between users, process control system and information management system.

The station shall have the capability to split the entire plant or areas of plant display into logical windows. Any graphic display at a VDU on the Operator Station may be changed, the changed graphic shall be able to be transferred/display to the other VDU on the Operator station without interference to the DCS.

Failure of any component of the system shall not limit the capability of the operator console to communicate with the system. For maximum availability, the client server configuration shall not be proposed.

Each workstation shall have a keyboard, and a mouse, track ball or light pen, which will be the operator's primary means of navigating the control system. The operator must be able to perform all the required system and process control functions described elsewhere in this specification. The controls on the operator console shall be physically separated according to their functions. The layout of the keyboard shall be designed such that a minimum number of operator actions are necessary to initiate a function.

a) Visual Display Unit (VDU)

From each colour VDU, the operator must be able to monitor all system and process functions. From any VDU the operator must be able to observe all parameters/displays.

The displays shall be of high resolution. The stability shall be such as to minimise requirements for colour alignment. The screen update shall be as below with immediate feedback of operator's action.

i) Acknowledge of Operator's Keystrokes

Acknowledgement of the operator's command from all key strokes shall take less than 100 ms (light, sound, etc.) with flashing "Wait" or other signal for the balance of the less than 2 seconds (see below).

ii) Display Up-Date Time

The following maximum faceplate display up-date times at the VDU shall be provided from any variable on the LAN:

- Routine up-date – ≤ 2.0 seconds
- update on command – ≤ 2.0 seconds
- change of state at the process – ≤ 2.0 seconds
- execute operator's command for all discrete outputs – ≤ 2.0 seconds
- controller set point change – ≤ 2.0 seconds
- for graphics, the graphic picture shall be completely drawn in less than 2.0 seconds while the process variable display "fill-up" may extend to a maximum of a further 1 second.

iii) Electronics

The electronics shall be low dissipation in state-of-the-art packaging without fans.

iv) Disk Drives

Floppy, CD/worm and hard drives shall be provided as required.

b) Operator's Work Station

The operator terminal shall be designed with a hierarchical format to minimise the number of steps performed by the operator to request a function. The operations required when requesting the functions shall be simple and logical and shall operate in a conversational mode. Whenever a program is selected, the system shall request data, accept the input and validate it before proceeding to the next item. Invalid operator input will cause an error response via a pop-up window on the VDU screen.

The operator's work station shall provide the means by which the operator accomplishes the task of monitoring and manipulating the plant. The basic functions provided by the work station shall include:

- Total plant hierarchical status overview and control
- Group overview
- Single loop and point display
- Interactive colour process graphics that are configurable by the user. They are to be of sufficient quantity and quality to enable the operator to control all process functions directly from the graphic displays ("live" graphics)
- Trending of all continuous variables.
- Historical record keeping – archival storage of variables to be stored on disc and then backed up on worm drive. Current data storage is to continue while archival data is being retrieved.
- Display control loop variables and discrete (one/off) inputs (via standard face plates).
- Alarm annunciation, display and reporting
- Logging of operator interventions to printer
- Video screen output to graphics printer
- Display/print shift, periodic and trip reports
- Manipulation of closed loop controllers, i.e. set points, control modes and control outputs
- Manipulation of discrete (binary) outputs
- Display of the status of hardware and interface equipment
- Drive interlock display
- Group Start/Stop display
- X-Y plots

c) Displays

The video display system shall have the capability of linking the different types of displays to move from one type to another and shall have the capability with split screens of viewing more than two types of displays at the same time on one VDU.

Video displays shall include process monitoring displays, alarm monitoring displays, Operator assistance instructions, system and subsystem graphics, and programmer/system data base changes described below:

(i) *Menu Display*

A structured menu display shall be included as the prime display for the plant. This display shall branch for various plant areas, with sub-branches for each plant item and further branches for actual displays.

(ii) *Overview Display*

The displays shall present the operator with the general overview of the process and I/Os and enable the operator to call up any graphic display by single selection. The graphic display shall also provide the Operator with an overview of individual deviations between the process functions and their assigned setpoint values.

(iii) *Operation instruction display*

Operation instruction display shall be designed to send messages/ instructions to the operator regarding the plant operation. The message shall support two functions; A message that notify the operator of the process progress and operation guidance. And dialog message that prompt the operator to acknowledge or input a value, and notify the control algorithm of the input results.

(iv) *Group Display*

The group display shall present the condition of the group of selected analogue or digital functions to the Operator. Each group shall have an address number and title. The system shall have the capability for all functions in the system, analogue or digital, to be configured in any group and displayed in as many groups as required. Additionally, the Operator shall be able to select any of the displayed control loops for control action, manipulate the control mode of the loop selected; change setpoint value; make bias adjustment; or initiate trending or hourly averages and other displays. For critical actions, a confirmation of the action will be required.

(v) *Detail Display*

This type of display shall show to the Operator a view of the condition of one process function while adjustments and changes are being made. The Operator shall be able to address any analogue or digital function in the system. Through this display, the Operator shall be able to manipulate the various modes of any loop as required in optimising controls. Changing of constants, tuning, zero and span value, alarm limits, control algorithm and other loop parameters shall be allowed or disallowed only through the use of password control.

Where there are interconnections, tabs shall be provided which allows the operator to select associated displays. One of these displays shall always be the main menu.

(vi) *P&ID Graphic Display*

This graphic display shall show the position of all active elements (e.g. CBs, valves) on the P&ID, and shall continuously update relevant analogue values. By use of associated Faceplate Displays which “pop-up” when a plant item is selected, the Operator shall be able to manipulate the control mode of that item. This shall include control loops (change setpoint value; make bias adjustment; and manipulate the output of a loop in manual) pumps (start, stop) and isolating valves (open, close).

The system display shall be capable to provide at least 200 kinds of dynamic mimic displays with up to 1000 variables.

The graphic display shall support dynamic graphic display symbols, dynamic data and links to the other displays and/or dynamic bar

It shall be possible to pan and zoom within this display to display increased details.

(vii) *Faceplate Display*

Faceplate displays shall utilise real-time database information. Faceplate displays shall be developed with the Principal for common control types, e.g. hand/auto stations, sequence control stations.

(viii) *Hourly Averages Display*

The last ten hourly averages of a maximum of eight process functions shall appear in tabular log-type format. All values shall be in engineering units and shall be available for any function from any controller. As a minimum the system shall have the provision for 120 hourly average displays.

(ix) *Trend Display*

The system shall have the capability to display graphics of historical trend data for process functions optionally selected from the group display. Trending shall be updated automatically with real time values. The Operator shall have the flexibility to select any time basis for trending.

The system display shall provide sixteen trend data of different sampling rate in one window.

It shall be possible to pan forward (to current time) or backward in time to look at past occurrences.

(x) *X-Y Plots*

The detailed requirements for the X-Y Plots are given in the Plant Management System. See Part 4 Section 3 I-200 “Plant Management System”

(xi) *Communication System Status Display*

This feature shall monitor all equipment on the data/communication networks, and shall warn the Operator of any malfunction. This display shall pin-point the location and define the cause of the problem.

(xii) *Alarming*

Analogue and digital functions shall be monitored in the following manner: High and low absolute value of an analogue function or high and low deviation from setpoint value, excessive rate-of-change of analogue point functions, and alarm on non target state or mismatch of output command for a digital point.

(xiii) *Alarm Summary Display*

This display shall warn the Operator whenever points are in Alarm State. The following information shall appear on the display: Time occurrence, scan status, point ID, alphanumeric description, units, current value, and alarm limit. The system shall be configured chronologically wherein the most current alarm shall have priority residence on top of the display, with time of occurrence displayed in hours, minutes and seconds. A point in alarm state shall be characterised by an audible sound and flashing on the display until acknowledgement. In the event the point returns to normal before acknowledgement, it will remain on the unacknowledged sub-screen display until acknowledged by the operator. The alarm silence pushbuttons and the alarm horn shall be located at the operator console (softkey). Alarms shall only be able to be acknowledged one at a time. In addition, a line shall be displayed on whatever other display is on view to inform the operator that an unacknowledged alarm is present. Positioning and energising the mouse on this line shall automatically call up the alarm summary screen.

(xiv) *Sequence Monitoring Display*

The detailed function of the Sequence Monitor is given in the Plant Management System. See Part 4 Section 3 I-200 "Plant Management System"

5.7.2 **Operator Keyboard**

The contractor shall provide a standard QWERTY keyboard with each console for text and data entry. The function keys on this keyboard shall be used for display selection, audible annunciator, mode selection for analog type set point and miscellaneous digital inputs for the operator stations. The operator keyboard shall be of the spill-proof type.

Using dedicated keyboard function keys, the engineer shall be able to perform with a single keystroke such actions as but not limited to the following:

- a) Alarm acknowledge
- b) Print an image of the current display including graphic

- c) Log and trend selection/assignment
- d) Report initiation
- e) Call up display
- f) Data entry
- g) Call up the prior display
- h) Page forward
- i) Page backward
- j) Display forward
- k) Display backward
- l) Call associated display
- m) Access a help display associated with the current display
- n) Access the message summary display
- o) Access the alarm summary display
- p) Access the alarm annunciation display
- q) Access up to 200 dedicated display
- r) Access to system menu

Historical trend display shall be presented in the same format as the real-time trends and shall enable to display at least one month of data from each process I/O name at the sample resolution of one minute per I/O name.

The display system shall provide the facility to configure a quick reference table to call up a group of frequently used displays

5.8 Plant Management System

Detail requirement for Plant Management System shall refer to Part 4 Technical requirement, Section 3 Technical Specifications I-200 “Plant Monitoring System”.

5.9 Engineer and Diagnostic System

Initialization and maintenance of all aspects of the data acquisition and control system shall be provided at the Engineering and Diagnostic Station. The system shall be completely self-sufficient and capable of editing existing programs and data files, building new functions and reinitializing and loading all control stations in the system while the system is in operation. The changed configuration shall only affect the related loops that are being changed.

The engineering and diagnostic system is a part of the DCS that is directly connected to the main data bus of the plant control system that shall be operated from a workstation based located in the Engineer’s room in the Central Control Building.

The main purposes of the engineering and diagnostic system shall be to perform the following:

- Engineering and data documentation (for commissioning and modification updating)
- On-line self-diagnostic routines for Maintenance and service for failed control system function.

The engineering and diagnostic system shall be designed as a single engineering system that enables the engineer to access all system configurations with uniform user graphic interface.

This system shall be provided with the following software:

- Writing program for solving special technological problems.
- Configuration of the automation systems on the basis of function diagrams.
- Configuration of bus communication
- Configuration of process control and information function
- Configuration of the hardware including parts lists, circuit diagrams, cable lists, etc.
- Generation of the entire plant documentation
- Monitoring the load of Plant Control System
- Monitoring the load of data high way
- Supporting start-up and maintenance tasks
- Fault diagnostic of all instrumentation and control components such as:
 - a) Unit Operation and Monitoring System
 - b) Plant Management System
 - c) Unit Engineering and Diagnostic System
 - d) Process Automation System
 - e) Bus System
 - f) Complete diagnostic function
 - g) Condensing I & C messages into overview displays
 - h) Interactive localization of source of fault with graphic representation
 - i) Detailed information with fault evaluation

If the tasks of Engineering and Diagnostic procedures cannot be supplied with only one system, one separate system for engineering and a second system for diagnostic purposes will be accepted.

The Engineering and Diagnostic Station shall also be provided with the ability to maintain the database of the plant's process and monitor plant performance. Process variables shall be created, deleted, modified or calibrated using this console. All changes shall be saved on bulk memory and/or documented for future use.

In addition to the above the following operations shall also be available in engineering mode of operation, selectable via a password:

- Database configuration
- Building custom graphics and displays
- Programming and compilation (procedural language programs via CAD engineering tool)
- Down loading (either partial or full code capability) of system configuration data and program to controller modes on LAN
- Controller tuning screen page
- On line programming facilities
- Change alarm set points
- Change controller tuning data including proportional, integral, derivative and other control block parameters
- Set time and date.
- Bypass criteria.
- Simulation.
- System analysis.

5.9.1 Engineer Workstation

Two Operator Stations shall be provided with the following requirements as minimum:

- Single 21" colour monitor.
- 1280 X 1024 pixel resolution.
- 256 distinct colours.
- brightness and contrast adjustment
- Screen build time within 2 (two) second.
- On-line screen refresh within 1 (one) second.
- Display 48 lines with minimum 80 characters per line.
- Adjustment density.
- Digital display adjustment

The Engineering Keyboard shall meet all requirements for general-purpose keys, and may in addition contain connection facilities for additional removable keyboards to expedite such function as system logic configuration.

The Engineering keyboard shall feature a security lock (password required) to limit access to set and engineering functions and shall be membrane-protected type or better.

5.9.2 Engineer System

The engineer shall be able to access detailed loop connections in graphical format for all configured tags in the system and the graphical format shall be an exact replica of the displays used for database configuration with real-time data indication and graphic templates.

A detailed display for sequence function blocks shall also be provided. The engineer shall be able to monitor all sequences logic in real-time using the same configuration formats that are used to configure the logic (i.e. in the form of logic diagrams or logic tables or logic script language).

Other displays shall also be provided for displaying historical events and self diagnostics with means for data retrieving and filtering.

The workstation for documentation, programming and commissioning functions shall include the following requirements as minimum.

- All engineering functions shall be capable of being executed off-line on a personal computer and later transferred to the on-line Engineer and Diagnostic Station.
- Special high-level language processors and software shall be available to the engineer for database and control loop modification, graphic and report creation.
- Database management
- Control configuration and modification.
- Graphic display and modification.
- Group, report and history file installation.
- Calibration of analog input channels and curve fit modification.
- Calculation of characteristic values and deviations from standard performance.
- Programming of Process Automatic Units via the Main Data Bus transmission system.
- Alteration of programming during operation (on-line optimizing)
- Simulation of input and output signals, of intermediate flags and times that shall be displayed on the plant control and monitoring system.
- Display of any desired signals on the bus transmission system (living function chart)

- Disturbances analysis and hierarchical fault detection for the control components.
- Back documentation (as built) of all modifications done by the engineer.

5.9.3 Diagnostic System

Each processing unit shall have the capability to periodically conduct a series of on-line self diagnostic routines in order to confirm the integrity of the circuitry and programming or identify the area and source of faults. LEDs visible from the front of the module shall indicate any such faults. An alarm shall then be initiated to allow the faulty component, circuit, sub system, etc. to be replaced.

The self-diagnostic shall comprise in minimum the following error check:

a) Bus structure:

Regular pattern check plus a power-up confidence check

b) Memory:

Regular pattern check on all memory areas and check sum on all user programming, configuration tables and executive memory

c) Power supply:

Have circuitry to monitor the normal state and report to executive memory via status register.

A Comprehensive set of diagnostic programs shall be supplied by the Contractor covering all hardware components supplied for DCS. This includes the microprocessor based controllers, peripherals, communication hardware modules, man-machine interface devices and any other hardware that is controlled by or communicates with the DCS.

The diagnostics functions shall permit, to the maximum possible extent, the identification and location of any hardware faults without requiring additional test equipment such as oscilloscope, etc. The repair philosophy shall be centered around “board” and unit replacement therefore the system faults shall be identifiable to the card level. No special training shall be required to perform the diagnostic’s function other than what was offered by the Contractor upon Bid submission.

The diagnostic strategy shall be fully hierarchical. No step of diagnostic shall use any capability not verified in a previous step. The diagnostics shall take full advantage of error-checking hardware. The diagnostic shall begin with built-in hardware tests which verify the

proper basic functions, the memory area required to execute diagnostics and the diagnostic's software load path.

Provision shall be made to enable the engineer to immediately view the specific error message in a single keystroke without going through a diagnostic display hierarchy.

Software diagnostic tests shall include both individual element and complete peripheral devices. It shall operate automatically, after initiation by the maintenance engineer, and report the status of each test to the programmer's console. Low-level diagnostic to allow checking of Sub-components is desirable. Diagnostics shall have sufficient test loops to allow diagnostic testing with an oscilloscope, when desired.

The Contractor's Bid proposal shall describe briefly the offered scope of diagnostic hardware and software. The description shall identify the devices that shall be used for input and output and the type of faults/problems that can be diagnosed.

The operation status of the ICMS items shall be available on diagnostic and maintenance display. All the information shall be available on real-time basis.

Diagnostic display shall include:

- System status display to show current status of all ICMS items and nodes.
- All hardware including I/O and system processors.
- Communication interface network

5.9.4 Portable Engineer Station (for PLC)

A portable engineering station shall be included the following devices but not be limited to:

- (a) Programming unit including data storage unit.
- (b) All required interface modules for the programmable controller itself.
- (c) All required cable connectors and cables to interconnect the EWS.

The portable engineering station software shall be capable of conducting both on-line and off-line control system for multiple controllers.

The detailed of portable engineering station shall be subject to approval by the Owner.

5.10 Power Supply and Grounding

Each electronic cubicle shall have its own power supply unit fed from two separate redundant power sources. The power supply units shall be of plug-in type and equipped with over voltage protection and external noise filter.

Separate redundant internal power supplies shall be provided for the controllers and the I/O modules.

The programmable controllers shall be operable through momentary dips or outages in electrical supply of up to twenty (20) milliseconds duration.

All equipment shall be connected to the common grounding system of the entire power plant.

5.11 Application Software

An important part of controller plant engineering is its comprehensive scope of applications addressing the particular needs of power plants. These applications shall be based on in-depth knowledge of and experience in the power generation industry, have been refined in close co-operation with clients, and shall be fully integrated into the control system.

The control system applications shall represent powerful tools for diagnosis, long-term monitoring and optimization of power plant processes. They shall range from detailed process history and logs to complex calculations and expert systems serving as diagnostic and decision support tools.

These applications shall provide refined and power tools for diagnosis, long-term monitoring and optimization of power plan process. The different packages shall be tailored to each specific plant. The comprehensive range of these packages shall include, e.g.,

- a detailed process history and logs to support extensive process analysis;
- performance calculation and optimization, where deviations of actual plant performance as compared with a model imply a sub-optimal operating mode or hint at maintenance;
- service life monitoring of critical power plant components such as boilers and turbines to facilitate predictive maintenance and extend equipment life; and
- energy management applications.

The application software configuration processing shall be through menu fill-in the blank configuration. The configuration shall be user friendly and provide help functions for each of the menu fields and extensive use of the built-in library.

The programming language and tools shall be used for all DCS equipments and shall conform to the relevant sections of IEC 61131 “Programmable Controllers”. The system configuration shall be able to be modified on-line

without shutting off the controller or put the controller in a configuration mode.

The modification shall not affect the other loops or interrupting to be process operation.

Control loops can be combined with motor control; start-up and shut-down sequences and fast interlocking logic, all in the same control program and using the same high-level function block oriented language software.

A controller shall be able to be configured in software fully on line, i.e. with the programs running and controlling the process. If required, a part of software program, a complete software program, a complete software program or the whole controller can be blocked during configuration. It shall also be possible to develop programs off-line in an engineering station and load them into the controller at a later stage.

In addition to functional PC elements, software shall contain a number of structural elements for division of a software program into suitable application program modules which can be managed and executed individually. The modules can be given different cycle-times and priorities so that both fast and slow control operations can be managed by the same software program.

The inputs and outputs of a PC element are connected to the inputs and outputs of other PC elements or to process I/O points, Picking these elements and making these connections shall constitute the configuration work,. The resulting software program can then be documented graphically.

The following documents shall be approved by the Owner before the software program is developed.

- Plant Control Logic
- Control Loop
- Graphic Display
- Description of System
- Instrumentation List

The individual document shall also be stored in CD-ROM to ensure that directories of chapter are maintained. This document shall include cover sheet, table of content, arrangement of plant component and include interface physical location, electrical overview, circuit diagram, terminal and termination diagram, list of setting values, all data electrical load (electrical consume list, measuring point list), etc as required.

The storage media for all data should be in CD-R and CD-RW. The DVD-RW itself shall have minimum capability to write at 52x, rewrites at 32x, and read at 52x or better.

5.11.1 Programming Facilities

Separate programming facilities shall be provided for each category or class of system provided by the Contractor. Programmer's consoles shall be freestanding suitable for locating in the technical staff room. Each console shall include a VDU screen, facilities for storing and loading programs in DVD-RW and a full function keyboard.

The contractor shall provide a special hardware and software for on line plant management system in order to improve the optimization of plant operational efficiency.

The contractor shall provide the owner with Licenses for all Software used in the plant for the operation life of the power plant, including the operating system software, so that the program supplied can be modified if required in the future.

5.11.2 Programming Methods

The DCS shall support a structured modular programming by providing the following standard operations:

- Logic functions (e.g. AND, OR, NAND, NOR, NOT etc)
- Timer functions (externally adjustable)
- Counter functions
- Skip functions
- Comparison functions
- Limit value functions
- Arithmetic functions
- Physical units
- Closed-loop functional such as P, PI, PD, PID, etc

The contractor shall submit the programming documentation for the Owner review and shall include the "As Built" documentation in the Instruction Manuals.

Software programs shall be developed utilizing the latest technology of programming and shall consist of, but not limited to following:

- Advanced Data And Alarm Management
- Multi-Level Password Security
- Efficient Memory And Data Manipulation
- Easy Directory Access
- Access To Engineering Workstation
- Form Data

- Help Function

All necessary software packages shall be provided along with the system. These software packages will include but not limited to the following:

- System Configuration Software
- System Management Software
- Workstation Access Configuration Software
- Real Time Display Software
- Graphic/Text Builder Software
- Historical Function Software
- Report Generation and Plant Management System Package
- Integrated Control Software
- Plant Optimization Software

5.11.3 Hierarchical Structure and Standard Functions

The application software shall be well structured with respect to functions and hierarchical levels.

The application software shall be structured in such a way to meet the technological process function's requirements that implemented in the DCS hierarchical structure.

Standard function elements shall be provided in software library and shall be consistently applied for the maximum possible extent.

6. TESTING

6.1 Factory Testing

Before shipment, the Contractor shall perform a complete functional test of the system including all peripherals and accessories at the factory using the final program and simulated inputs or simulated process if available. The Owner /Engineer shall have the opportunity to participate in the simulation and to introduce random upsets as desired. The details of the factory test program shall be provided with the Bid Proposal. Refer also to Part 4, E-002 "Test and inspection".

The Owner reserves the right to refuse shipment of the system until a satisfactory system test has been demonstrated.

6.2 Testing at Site

When the Contractor's equipment is put into service, the Contractor shall run an availability test run to ascertain that the equipment is performing properly in every respect. This shall be in addition to tests performed at laboratories and

Contractor's factory prior to shipment. Refer also to Part 4, section 4" Erection and Commissioning" Subsection 4.5" Instrumentation and Control".

The Contractor shall guarantee and successfully demonstrate that the Control & Monitoring Systems (including hardware and software) shall perform as specified with a total up-time in excess of 99.8% for a continuous period of 90 days (2160 hours) after commercial operation of the plant prior to taking over by the Owner. This demonstration is in Owner's plant with all input-output devices connected and all functions in service.

The demonstrated availability shall be calculated as follows

$$\text{- \%availability} = 100 (A - B) / A$$

A : Test duration time

B : Accumulated control and monitoring system outage time

Availability shall be defined as the proper functioning of the control & monitoring system components so that all system functions are performed and no output information is lost. Where a failure does not cause loss of a function, even though individual items within that function may be lost, such function does not count as downtime.

In the event that accumulated outages exceed the specified allowable outage in the demonstration period, the starting time of the availability test will be shifted forward to delete previous accumulated running time and downtime until there is a demonstration period in which the availability meets or exceeds the specified requirement.

Control & Monitoring System failure criteria shall be defined as follows

System Failure Criteria

Distributed Processing: Any failure such that the complete processing capability of that CPU is unavailable shall be considered as a system failure.

Plant Process I/O: The loss of any capability of Interface of processing plant signal inputs shall be defined as a system failure. In addition, the loss of more than 20 system inputs or outputs shall be defined as a system failure.

VDU Display Capability: The simultaneous loss of the capability to display information on more than any two VDU's (system wide) shall be considered a system failure.

Operator's Interface: The loss of any functional capability on two operator's keyboard, or both on operator's keyboard and programmer's/ engineer's keyboard shall be considered a system failure.

System Loggers: The simultaneous loss of full capability to print as specified by the alarm and periodic log printer shall be considered a system failure.

The Owner shall have the option of performing required maintenance during 90 days of the demonstration period (to be mutually agreed upon) without active supervision by the Contractor's personnel. If Owner's personnel require assistance to return system to operation after a malfunction, Contractor's local service engineer will be called. Contractor shall maintain capable service engineers at the Site on call 24 hours a day, seven days a week during the entire period of the availability run.

Following a system malfunction, the Owner will call the Contractor to inform him of the apparent nature of the malfunction. (If the Contractor cannot be reached, downtime will begin immediately.)

The downtime shall end when the system is returned to satisfactory service.

For purposes of this guarantee, the system shall be defined as including the microprocessor based control system, the data acquisition system, the input/output processing system, the data communication system, control stations, and all peripheral equipment furnished under this Specification.

Failure of an output printer shall not be considered as a system outage provided a spare printer is available.

Contract maintenance shall be provided by the Contractor until the end of availability run. During this period, the Contractor will perform the scheduled preventive maintenance on the peripherals and the system and the stock of Owner's spare parts will be available for the Contractor's use. However the contractor shall promptly replace any spare parts which he uses to repair the system during this period (at no additional cost to the Owner).

Contractor shall guarantee that the overall accuracy of the input system shall be as specified.

Contractor shall guarantee that the control and monitoring system shall be able to control all the inputs to the power plant, systems to maintain the output parameters of the power plant equipment within the limits specified herein.

The availability run will commence at the following time:

The availability run shall start on the scheduled date of the initial operation of the power plant unless it is mutually agreed by both Contractor and Owner that it shall start at an earlier date.

The availability run will not commence, if in the opinion of the Owner, the controls and monitoring system is not ready to start the availability run.

The Contractor shall be responsible to provide maintenance until the completion of the availability run.

Two times failure of the availability run shall constitute that the control & monitoring system did not meet the availability requirements and shall be ground for the Owner to reject the Contractor's control and monitoring system.

6.3 Test Equipment

Contractor shall furnish all test equipment or tools required for normal maintenance, testing and calibration. In addition, a separate memory configuration device shall be included in the proposal. This device shall be complete with all necessary appurtenances to permit rapid reconfiguration of the memory.

7. WARRANTY

The following hardware and software documentations shall be provided but not be limited to:

- (a) Block diagram
- (b) Schematic diagram
- (c) I/O list
- (d) Interconnection wiring diagram
- (e) Application software diagram
- (f) Graphic diagram for VDU display

The detailed documentation lists shall be subject to approval by the Owner.

The documentation requirements shall meet with the technical specification in Part 4, Sub-section 2.2 "Data and Drawing Submission".

8. QUALITY ASSURANCE / QUALITY CONTROL (QA/QC)

Quality assurance/quality control shall conform to the requirements of Part 4.Subsection 2.7.

The quality control procedures as specified in above Subsection 2.7, shall include but not be limited to the following

- Procurement
- Assembly

- Programming
- Testing
- Packaging and shipping

The quality assurance/quality control documentation as specified in above Subsection 2.7, shall include but not be limited to the following:

- Shop test reports
- All other documentation required by applicable codes and standards.

9. SPARE PARTS

20% of each type of modules shall be provided.

One complete required size rack shall be provided for housing the spare modules and for testing purpose.

(Last Page I-005)